

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of B. Sc. (Biological Sciences) Backlog Examination, September 2018

Course code & Name: BS 210 & Analytical Techniques in Biology

Date: 19/09/2018 Day: Wednesday Time: 10:00 A.M. To 10:30 A.M. Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I	Choose the correct answer for the following questions.		20
1.	Ion exchange chromatography separates biomolecules based on their _____.		01
	(i) Net surface charge	(ii) Affinity for bound ligand	
	(iii) Molecular weight	(iv) None of them	
2.	The following electromagnetic radiations have longer wavelengths than visible light.		01
	(i) Infrared radiation	(ii) UV-rays	
	(iii) X-rays	(iv) All of them	
3.	The following method is used for trans-blotting proteins separated by SDS-PAGE to a nitrocellulose membrane for their identification using antibodies.		01
	(i) Southern blotting	(ii) Eastern blotting	
	(iii) Western blotting	(iv) Northern blotting	
4.	Of the following which component is required for a polymerase chain reaction (PCR).		01
	(i) Taq DNA polymerase	(ii) Primers	
	(iii) dNTPs	(iv) All of them	

5.	The sedimentation rate of a given biological particle will be _____ when the density of the particle and the surrounding medium are equal.		01
	(i) Zero	(ii) One	
	(iii) Two	(iv) Three	
6.	In a column chromatographic separation when a single solvent (mobile phase) is used as an eluant the process is known as		01
	(i) Gradient elution	(ii) Isopycnic elution	
	(iii) Isocratic elution	(iv) All of them	
7.	The following enzyme is used to break peptidoglycan layer in the bacterial cell wall for isolation of genomic DNA.		01
	(i) Lysozyme	(ii) Chitinase	
	(iii) Cellulase	(iv) All of them	
8.	In SDS-PAGE, the following reagent is used to reduce disulphide bridges that hold the protein tertiary structure together.		01
	(i) Ammonium persulfate	(ii) β -mercaptoethanol	
	(iii) TEMED	(iv) Glycerol	
9.	The following chromatography is based on the unique property of extremely specific biological interactions (such as antigen-antibody, protein-ligand etc.) to achieve analyte separation.		01
	(i) Ion-exchange	(ii) Gel-filtration	
	(iii) Affinity	(iv) All of them	
10.	In agarose gel electrophoresis, the following DNA intercalating dye is added to the agarose gels in order to visualize the DNA bands on a UV-transilluminator.		01
	(i) Bromophenol blue	(ii) Ethidium bromide	
	(iii) Blue dextran	(iv) EDTA	
11.	The following salt is the most commonly used for concentration of proteins by precipitation.		01
	(i) Phenol-chloroform	(ii) Ammonium sulphate	
	(iii) Ethanol	(iv) Polyacrylamide	
12.	In an epifluorescence microscope, _____ reflects light of shorter wavelengths (excitation light) but allows light of longer wavelengths to pass through (emitted light).		01
	(i) Mercury Arc Lamp	(ii) Phase ring	
	(iii) Electron beam	(iv) Dichromatic mirror	
13.	In the following spectroscopy instrument, the detector is placed perpendicular (90° arrangement) to the excitation pathway in order to reduce the background of the incident beam of radiation.		01
	(i) Spectrophotometer	(ii) Colorimeter	
	(iii) Spectrofluorimeter	(iv) All of them	

14.	Of the following, which microscope has highest resolution?		01
	(i) Light Microscope	(ii) Fluorescence Microscope	
	(iii) Electron	(iv) All of them	
15.	The following is NOT the component of a light microscope?		01
	(i) Objective lens	(ii) Electromagnetic lens	
	(iii) Condensor lens	(iv) Occular lens	
16.	The _____ law states that; $A = \epsilon c l$, where A =absorbance, ϵ = molar extinction coefficient, c = concentration, l = path length.		01
	(i) Beer-Lambert's Law	(ii) Charles law	
	(iii) Boyle's Law	(iv) Ohm's law	
17.	In DNA isolation, the following reagent is used to chelate divalent cations that are essential for cell membrane integrity and activity of nucleases.		01
	(i) Ethidium bromide	(ii) Proteinase K	
	(iii) EDTA	(iv) SDS	
18.	An ion-exchanger resin consisting of sulphonic acid groups ($-\text{SO}_3^-$) can be classified as a _____.		01
	(i) Strong cation exchanger	(ii) Weak cation exchanger	
	(iii) Strong anion exchanger	(iv) Weak anion exchanger	
19.	In Thin-Layer chromatography _____ is used as the stationary phase		01
	(i) Whatman paper	(ii) Solvent	
	(iii) Silica gel	(iv) None of them	
20.	Two types of molecular vibrations, stretching and bending occur upon absorption of _____ radiations by molecules with unsymmetrical charge distribution.		01
	(i) Visible	(ii) Ultraviolet	
	(iii) Infrared	(iv) All of them	

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Course code & Name: BS 210 & Analytical Techniques in Biology

Date: 19/09/2018 Day: Wednesday Time: 10:30 A.M. To 1:00 P.M. Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION - I		
Q – II Answer the following questions as directed (answer any four).		20
1.	Explain the ascending, descending and 2-D paper chromatography methods.	05
2.	Explain the principle of following: (a) Gel-filtration chromatography (b) Affinity chromatography	05
3.	Describe the cyclic steps of a polymerase chain reaction (PCR).	05
4.	Explain the southern blotting method.	05
5.	Explain the preparation of total DNA from bacterial cells.	05
6.	Write a short note on protein purification.	05
SECTION - II		
Q-III Answer the following questions as directed (answer any six).		30
1.	Explain the working of a light microscope.	05
2.	Explain the alkaline-lysis method for plasmid DNA isolation.	05
3.	Describe the instrumentation of an atomic absorption spectrophotometer.	05
4.	Define the following terms: (a) Distribution co-efficient (K_d) (b) Retardation factor (R_f)	05
5.	Explain the most common types of rotors used in biochemical centrifugation.	05
6.	Explain the optical arrangements and working of a phase contrast microscope.	05
7.	Describe the instrumentation of a dual-beam UV/Visible spectrophotometer	05
8.	Write a short note on agarose gel electrophoresis.	05